In-vitro Biology

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Biochemical Assays

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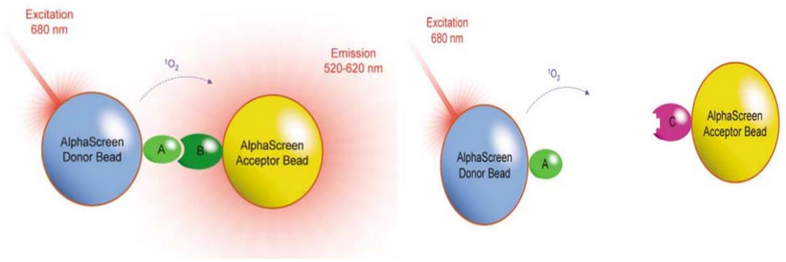
Case study: AlphaScreen

Assay principle:



High signal **No signal**

Brief protocol:

3 µl of Compound/DMSO + 3 µl of Enzyme (His tagged)

60 mins at 25 ° C

3 µl of of GTPγS + 3 µl of effector(GST tagged)

60 mins at 25 ° C

3 µl of bead mixture (Alpha glutathione donor beads and Anti-6xHis acceptor beads)

60 mins at 25 ° C

Read on Envision
Ex- Laser based; Em – 570 nm

Assay Optimization

A :Checkerboard to determine enzyme and substrate concentration

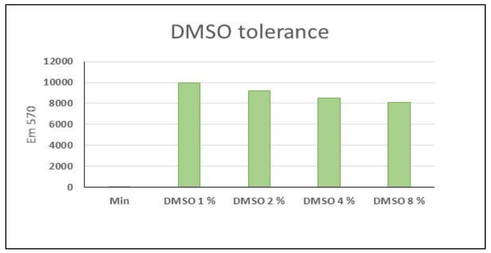
Checkerboard		Enzyme Conc (nM)								
		320	160	80	40	20	10	5	2.5	0
Substrate Conc (nM)	320	7485	6539	8497	8329	5507	5022	3169	1025	158
	160	6691	9819	11438	9590	7217	7531	3349	1626	127
	80	10872	10450	13858	11814	11990	8467	4846	2272	179
	40	9519	10864	12356	12656	8815	8365	4752	2100	139
	20	10186	11716	13669	12237	10323	6760	3627	1564	171
	10	5361	9810	10185	8248	6645	4568	2093	797	153
	5	3202	6381	6513	6591	4628	2582	1201	413	150
	2.5	1828	3223	3237	3526	2594	1293	550	255	132
	0	301	231	137	156	154	130	150	186	130

From the optimized enzyme and substrate concentration we performed a DMSO tolerance followed by compound validation

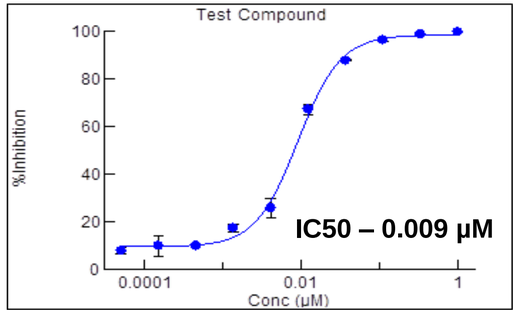
B : Buffer optimization

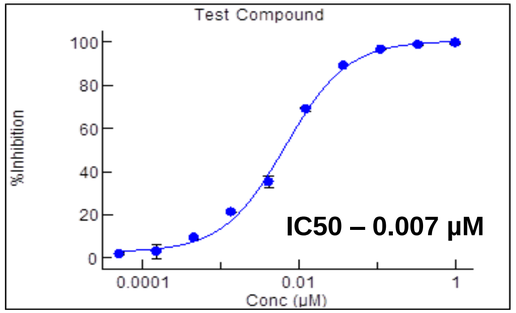
EDTA/Mg2+ Checkerboard	Max (GTPγS)	EDTA 8 mM	EDTA 4 mM	EDTA 2 mM
Mg2+	8 mM	1308	491	546
	4 mM	6062	**7610**	349
	2 mM	3515	6147	1599

C : DMSO Tolerance



D : Validation with tool compound





Quantification:
Luminescence
Instrument used:
Envision 2105
QC parameters used for assay:
a) % CV : < 20 %
b) Z' > 0.5
C) Signal Window : 30

This AlpaLISA assay was used to screen ~160 compounds for an ongoing IDD program. Proteins were made in-house, and the compounds were synthesized and provided by the APSL chemistry team.

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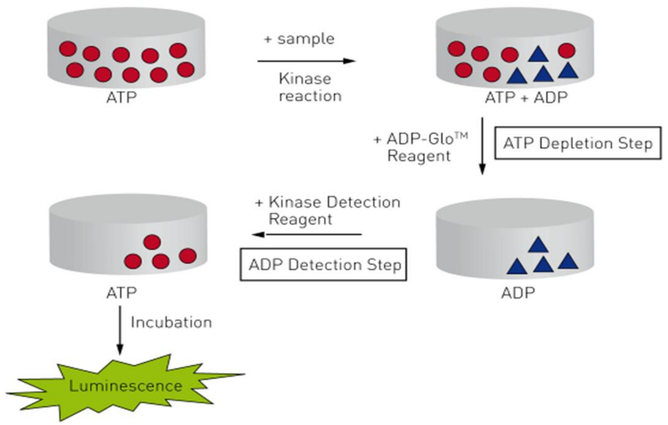
Case study: Kinase Assay using ADP Glo

Assay principle:



Brief protocol:

2 µl of Enzyme Mix + 1 µl of Compound/DMSO

15 mins at 25 ° C

2 µl of substrate mix

60 mins at 25 ° C

5 µl of ADP Glo reagent

30 mins at 25 ° C

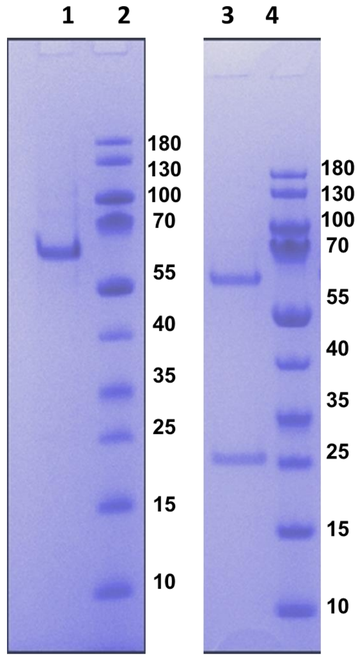
10 µl of Kinase detection reagent

30 mins at 25 ° C

Measurement of luminescence in Envision 2105

Kit Used- ***ADP-Glo Kinase Assay - Promega (V9101)***

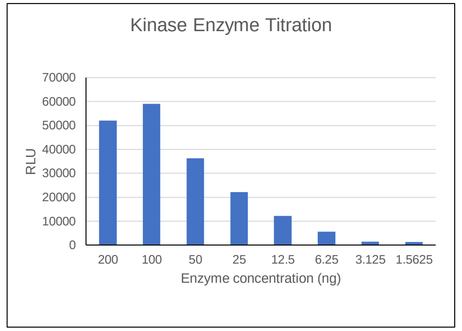
Protein



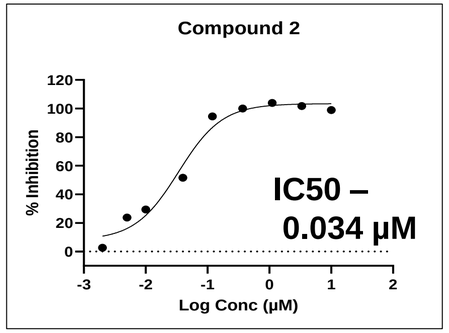
Gel lanes:

1. Inhouse protein
2. Protein Marker
3. Commercial protein
4. Protein Marker

Enzyme Titration



DRC of Compound



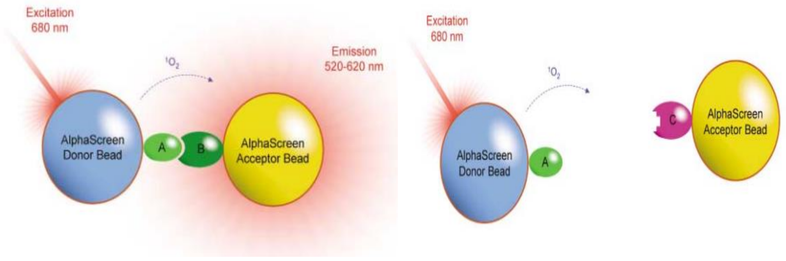
Quantification: Luminescence
Instrument used: Envision 2105
QC parameters used for assay:

a) % CV : < 20 %
b) Z' > 0.5
c) Signal Window : 18

- Initially commercial protein from Promega was used, later inhouse generated protein was used for screening.
- Over 200 novel compounds screened as part of an IDD programme.
- Inhouse protein comparable with commercial protein with better purity

Case study: AlphaScreen

Assay principle:



High signal **No signal**

Brief protocol:

3 µl of Compound/DMSO + 3 µl of Enzyme (His tagged)

60 mins at 25 °C

3 µl of of GTPγS + 3 µl of effector(GST tagged)

60 mins at 25 ° C

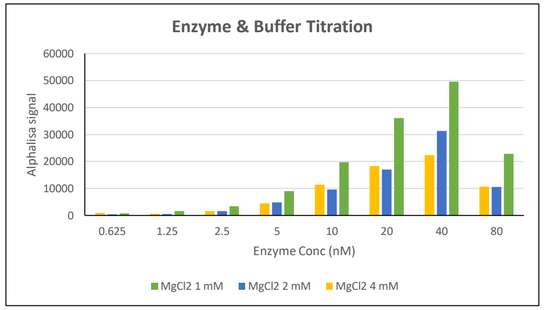
3 µl of bead mixture (Alpha glutathione donor beads and Anti-6xHis acceptor beads)

60 mins at 25 ° C

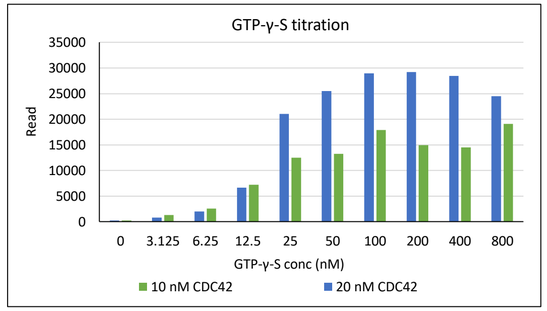
Read on Envision
Ex- Laser based; Em – 570 nm

Assay Optimization

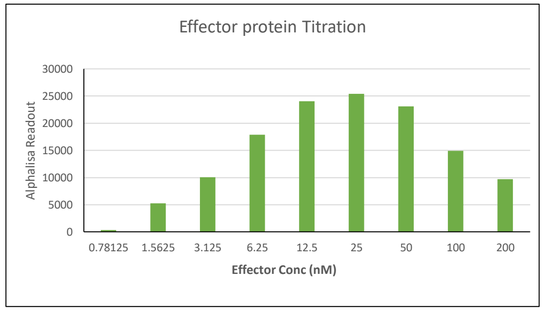
A : Enzyme & Buffer Titration



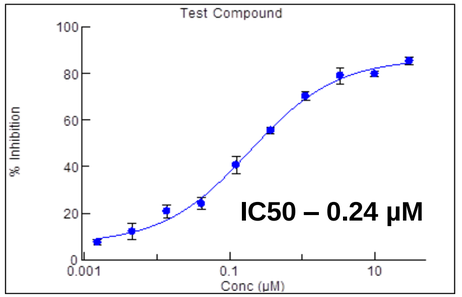
B : GTPγS Titration



C : Effector Titration



D : Validation with standard



Quantification: Luminescence
Instrument used: Envision 2105
QC parameters used for assay:
a) % CV : < 20 %
b) Z' > 0.5
C) Signal Window : 30

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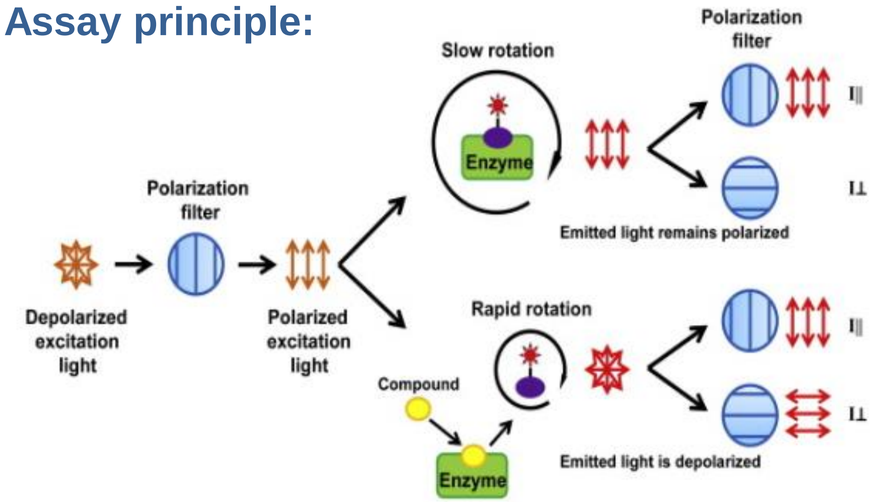
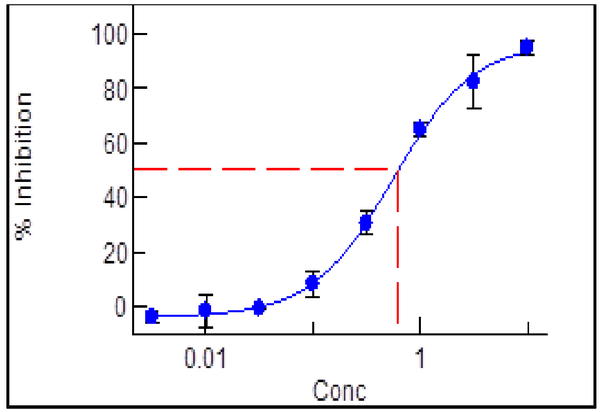
Case study: Binding assay - FP assay

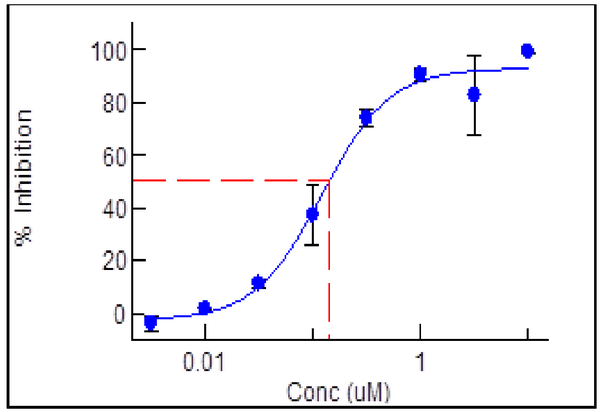
Assay principle:



Inhibition curve for reference compound-1



Inhibition curve for reference compound-2



Brief protocol:

2 µl of Compound/DMSO + 3 µl of protein

45 mins at 25C

2 µl of probe

45 mins at 25C

Read Fluorescence on Envision

- **Quantification:** Fluorescence
- **Instrument used:** Envision 2105 plate reader
- **Reference compounds:** commercially procured
- **QC parameters used for FP assay:**
  a) % CV : < 10 %
  b) Z' : > 0.5

Overview:

- Optimized and validated 384-well plate assay format for determining the competitive binding of test-compounds using FP assay
- Useful for screening inhibitors for protein-probe interaction
- APSL has its in-house HTS screening team for screening

Over 1900 compounds screened for ongoing Novartis programme using this assay technology. Protein, probe and prediluted DMSO compound plates were supplied by the client.

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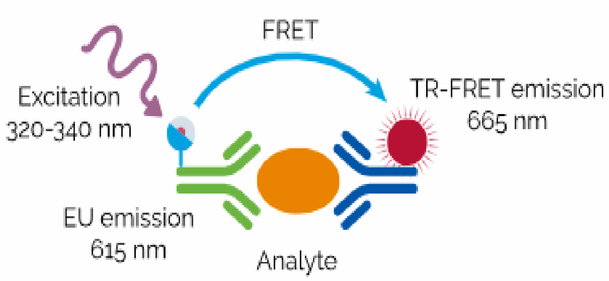
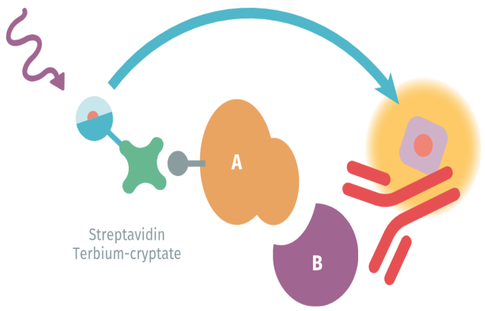
Case study: TR-FRET Biochemical Assay

Assay principle:





Brief Protocol:

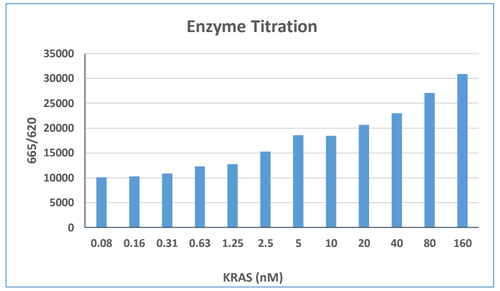
4 µl of Compound / DMSO , 4 µl of Enzyme (avi tagged),
4 µl of Tb-SA, 4 µl of Tracer

60 mins at 25 °C

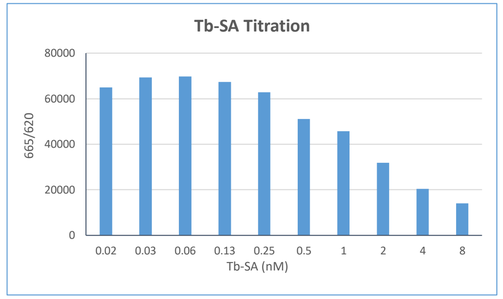
HTRF Measurement - TRF Laser
Ex- 340 nm
Em- 620 nm/ 665 nm

Assay Optimization

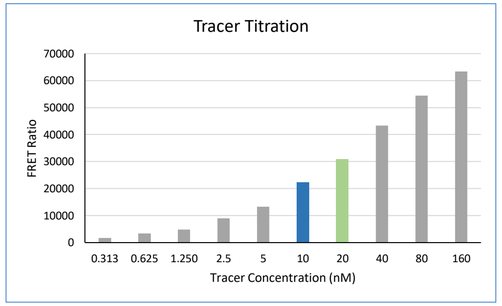
A : Enzyme Titration



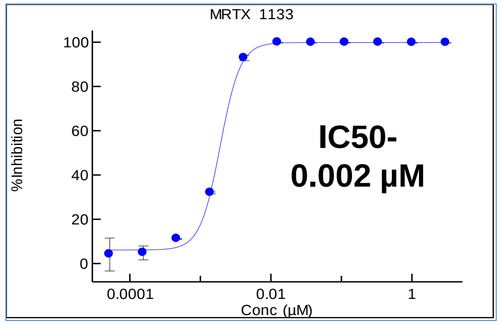
B : Tb-SA Titration



C : Tracer Titration



D : Validation with tool compound



Quantification: Luminescence
Instrument used: Envision 2105
QC parameters used for assay:
a) % CV : < 20 %
b) Z' : 0.5
c) Signal Window : ~40

Around 60 compounds screened for an FTE programme using this assay technology. Protein was supplied by client. Tracer and test compounds were made inhouse by chemistry team, Tb-SA cryptate was procured commercially.

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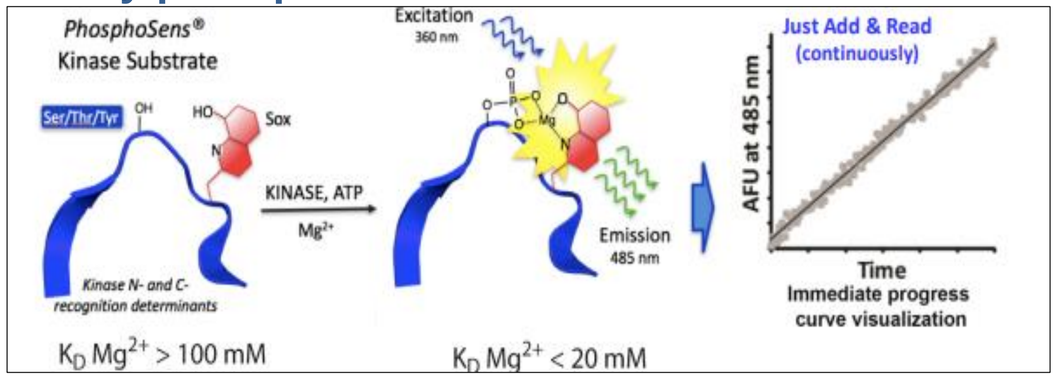
Case Study : AssayQuant kinase activity assay-

Assay principle:



Brief protocol:

Prepare master mix containing the PhosphoSens substrate as per the manufacturer's details

Transfer 5 µl of compound/ DMSO in assay plate and add 35 µl of master mix to each well

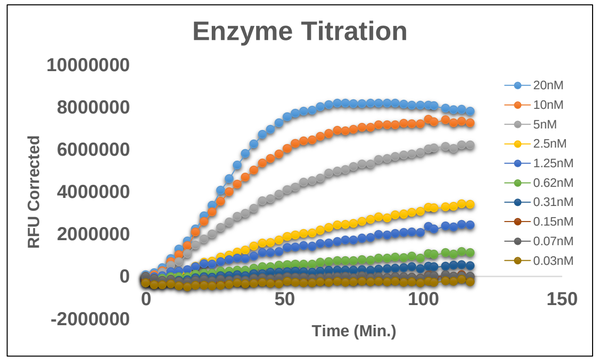
30 min at 25 °C

Measure fluorescence every 20 mins for 2 h
(Ex- 360 nm, Em – 492 nm)

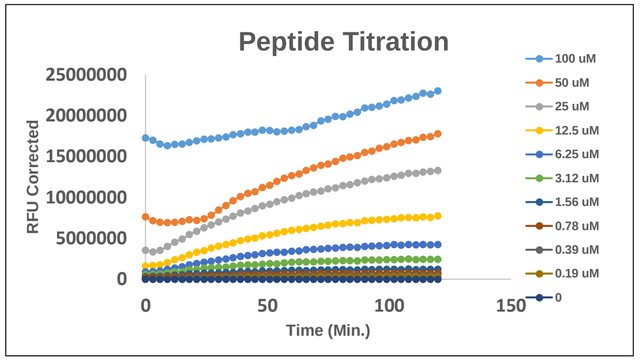
Kit Used- AssayQuant PhosphoSens

Assay Optimization

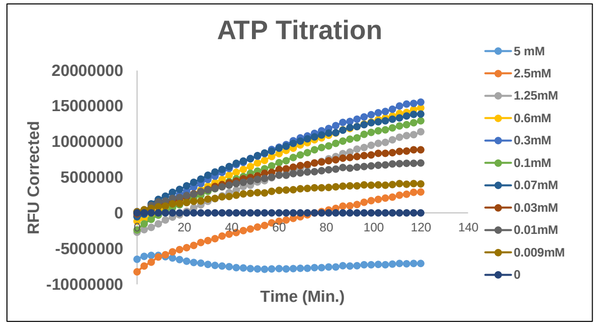
A : Kinase Titration



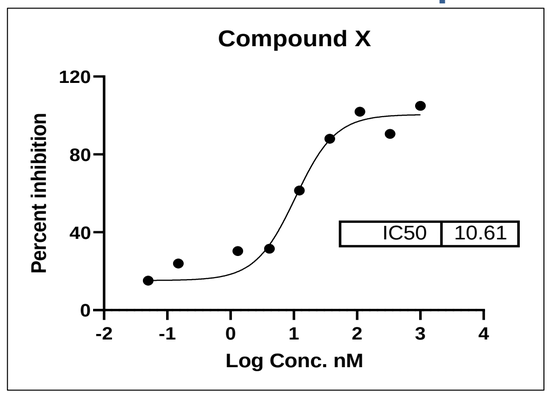
B : Peptide Titration



C : ATP Titration



D : Validation with compound



Quantification: Fluorescence
Instrument used: Envision 2105
QC parameters used for assay:
a) % CV : < 20 %
b) Z' : > 0.5

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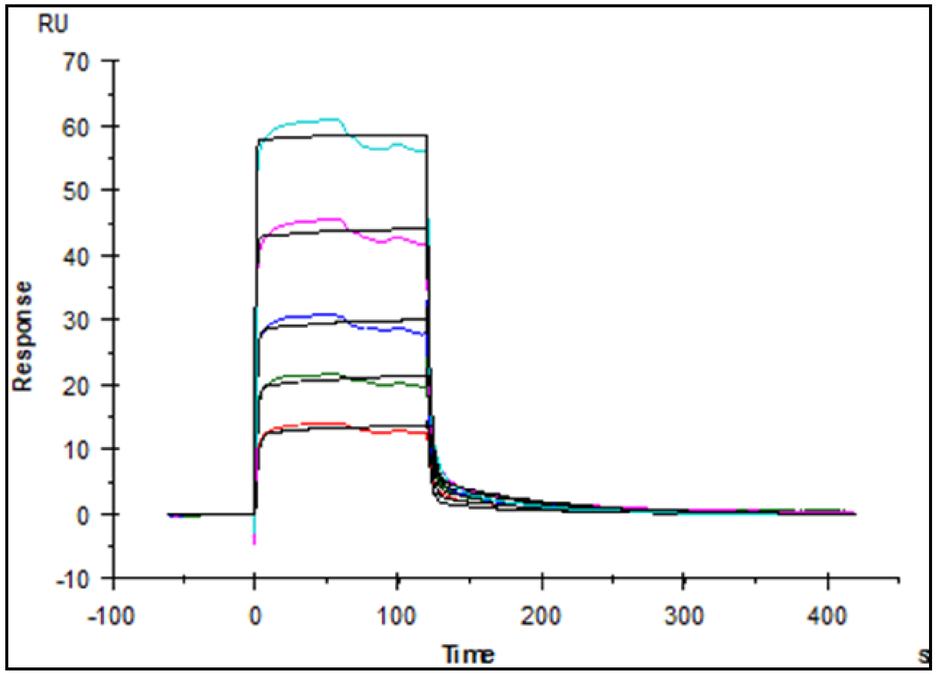
Kinetic binding analysis by SPR

Ligand Immobilization on CM5 sensor chip → **Kinetic binding analysis of test compounds**

Experimental parameters

Chip Type	CM5
Running Buffer	PBSP + 1% DMSO
Analyte	Contact time – 120 seconds, Dissociation time – 300 seconds, Flow rate - 30 µl/minute, Extra wash – 50% DMSO
Temperature	Sample Compartment – 25^{0}C, Analysis – 25^{0}C

Test Compound



Kinetic parameters (Two state reaction fit)					
Sample	**ka (1/Ms)**	**kd (1/s)**	**KD (µM)**	**Rmax (RU)**	**Chi² (RU²)**
Compound A	4.14E+04	0.3965	8.260	52.5	0.829

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Case Study 1: Compound screening (Assay developed at customer site adapted at APSL)

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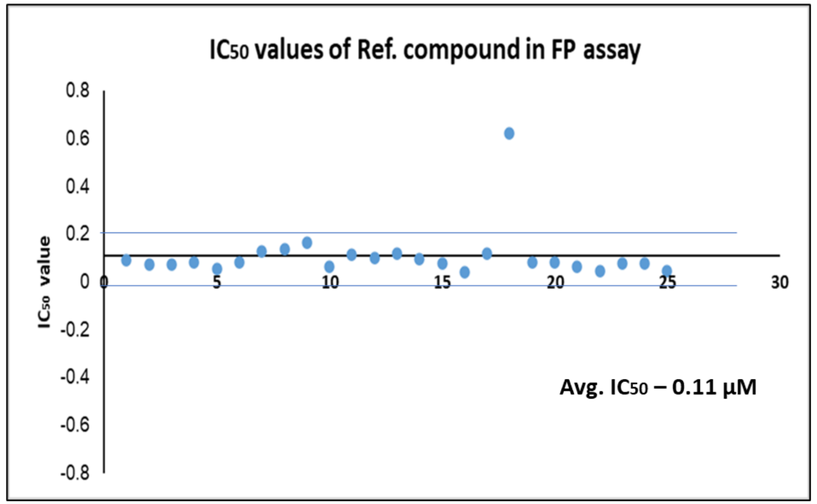
- **Adaptation and establishment**
  - **From 1536-plate format to 384-well plate format, as per customer need and match APSL infrastructure**
  - **Fluorescence Polarization (FP) assay – 8 point (duplicates in separate plates)**
  - **HiBiT assay: 3 targets, 8-point DRC (duplicates in separate plates)**

- **Phase-I establishment of QC parameters and acceptance criteria: ~600 compounds**
- **Phase-II Screening mode (manual): On-going for 18 months: ~250 compounds/month**
- **Currently transferring to HTS mode using Bravo liquid handler**



Day-0: Compd. received → Day-1: Compd. dilution & assay initiation

→ Day-1: FP assay → Day- 2: FP assay data analysis and upload at customer site

→ Day-2 and 3: HiBit Assay → Day- 4: HiBit assay data analysis and upload at customer site